

Course 3: Operations Research

A. OVERVIEW OF THE COURSE:

Name of the Course	Operations Research
Offering Department	Mathematics
B.Tech. Branches to which this course is offered	B.Tech. (Mathematics and Computing)

Overview:

This course on Operations Research offers a rigorous foundation in modeling and optimizing complex systems, essential in engineering, management, and decision sciences. It covers network models such as shortest-path, maximum-flow, CPM–PERT scheduling, and minimum-spanning trees. Students will explore Multi objective programming techniques, including goal programming, with real-life case studies. The course introduces dynamic programming using Bellman’s Optimality Principle, along with standard queuing models like M/M/1, M/M/1/c, and M/M/c for system performance analysis. It concludes with Monte Carlo simulation methods for modeling uncertainty and evaluating real-world scenarios.

B. Syllabus of the course

Course No.	Title of the Course	Course Structure	Pre-Requisite
CMMTE303	Operations Research	3L-0T-2P	None

COURSE OUTCOMES:

- 1: To formulate and solving some important network models
- 2: Understanding the real-life application, formulation and solution procedure for multi objective programs.
- 3: To learn the application and solution procedure for Dynamic programs.
- 4: To understand the basic Queuing systems.
- 5: To acquaint the students with the simulation techniques for important distributions.

COURSE CONTENT:

UNIT-I: Network models and project management: Introduction, shortest-path problems, maximum-flow problems, CPM–PERT project-scheduling models, and minimum-spanning tree problems.

UNIT-II: Multi objective programming: Introduction with real life examples, Efficient frontier, Weak efficient and properly efficient solutions, weighted sum approach, goal programming problems, case study showing the application of Mult objective programming in real life scenario.

UNIT-III: Dynamic programming: Discrete and Continuous Dynamic Programming, Bellman's Optimality Principle.

UNIT-IV: Queuing models: Introduction and basic definitions, Characteristics of a Queuing System, M/M/1, M/M/1/c, M/M/c models.

UNIT-V: Simulation: Basic terminology, random number and monte Carlo simulation, random number generator, simulations with continuous random variables: exponential, uniform, triangular and normal distribution, inverse transformation method, applications of simulation with practical examples.

List of some of the experiments:

1. Solving shortest-path problem
2. Solving maximum flow problem
3. Solving CPM/PERT models
4. Solving Mult objective program using weighted sum approach
5. Solving Mult objective program using epsilon constraint method
6. Solving Dynamic program
7. Finding models characteristics for M/M/1, M/M/1/c and M/M/S
8. Generating model parameters using simulation techniques
9. Solving Queuing models using simulations
10. Network models using simulations

Reference Books:

1. Hillier, F. S., & Liebermann, G. J. (2006). Introduction to operations research (8th ed.). New Delhi: Tata Mc Graw Hill.
2. Ravindran, A., Phillips, D.T. and Solberg, J.J., “Operations Research: Principles and Practice”, 2nd Edition, John Wiley and Sons, NY.
3. Pant, J.C.: “Introduction to Optimization/ Operations Research”, 7th Edition, Jain Brothers, New Delhi.
4. Sharma, J. K. (2009). Operations research: Theory and applications (4th ed.). New Delhi: Macmillan.
5. Srinivasan, G. (2007). Operations research: Principles and applications. New Delhi: Prentice Hall of India.
6. Winston, W. L. (2009). Operations research: Applications and algorithms (4th ed.). Delhi: Cengage Learning.
7. Albright, S. C., & Winston, W. L. (2009). Management science: Modelling with spreadsheets. Delhi: Cengage Learning.

C. LESSON PLAN for THEORY LECTURE

Lecture No.	Topic to be covered	Unit
1	Introduction to network models and operations research applications	Unit-I (8)
2	Shortest-path problems	
3	Maximum-flow problems	
4	Applications of maximum-flow models	
5	Minimum-spanning tree problems	
6	Introduction to project scheduling, CPM (Critical Path Method)	
7	PERT (Program Evaluation and Review Technique) and comparison with CPM	
8	Case studies and applications in network/project management	
9	Introduction to Mult objective programming and real-life examples	Unit-II (8)
10	Concept of efficiency and pareto optimality	
11	Types of efficient solutions	
12	Weighted Sum Approach	
13	Goal Programming – Introduction	
14	Goal Programming – Solution Techniques	
15	Case Study: Real-Life Application of MOP	
16	Review and Practice Session	
MID SEMESTER EXAMS		
17	Introduction to dynamic programming	Unit-III (7)
18	Discrete dynamic programming – formulation	
19	Applications of discrete dynamic programming	
20	Continuous dynamic programming	
21	Bellman's Optimality Principle	
22	Multi-stage decision problems	
23	Case study/examples using dynamic programming	
24	Introduction to queuing theory and basic definitions	Unit-IV (7)
25	Characteristics and components of queuing systems	
26	M/M/1 queuing model – assumptions, steady-state analysis	
27	M/M/1/c finite capacity model	
28	M/M/c multi server model	
29	Performance metrics: average queue length, waiting time, etc.	
30	Applications of queuing models in engineering and service systems	
31	Introduction to simulation and Monte Carlo methods	Unit-V (10)
32	Random number generation techniques	
33	Inverse transformation method	
34	Simulation using exponential distribution	
35	Simulation using uniform and triangular distributions	
36	Simulation using normal distribution	
37	Hands-on problem: single-server simulation	
38	Simulation case study 1: production/service systems	
39	Simulation case study 2: inventory/queue systems	
40	Summary, integration of methods, and review	

